Aspiration Pump



Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Aspiration pump (peristaltic or vaccum)

What it is used for

An aspiration pump removes fluids, mucus, or other secretions through a suction system for clinical or laboratory use, depending on the device design. Operate it only for its intended purpose and follow the manufacturer's instructions, infection-control procedures, and the relevant clinical protocol.

Main parts

- Pump unit and motor
- Vacuum regulator and pressure gauge or display
- Collection canister or fluid trap
- Suction tubing and patient-side connector, where applicable
- Bacterial or overflow filter, where fitted
- Power supply, control switch, and optional foot switch

Before you start

1. Confirm the device type, intended application, and authorised procedure; patient use must be handled by trained healthcare personnel.
2. Inspect the pump housing, power cable, controls, vacuum gauge, canister, tubing, and connectors for damage.
3. Check that the collection canister is clean, correctly fitted, and below its maximum fill line.
4. Confirm that the correct filter and approved single-use or reprocessed accessories are installed as specified.
5. Verify the electrical supply and any required alarm or battery status before use.
6. For patient care, confirm the clinical order, required suction setting, patient identity, and infection-control precautions.

How to use

1. Place the pump on a stable, dry surface with ventilation openings unobstructed.
2. Connect the collection canister, filter, and tubing according to the device diagram.
3. Confirm that tubing connections are secure and that the fluid trap or overflow protection is correctly fitted.
4. Switch on the pump and check that the vacuum indicator responds normally.
5. Set the suction pressure only to the level prescribed by the authorised clinical protocol or specified procedure.

6. For patient use, trained personnel should explain the procedure, position the patient appropriately, and connect the approved patient interface using aseptic technique as required.
7. Apply suction only according to the relevant clinical protocol and the device's intended use; continuously observe the patient where applicable.
8. Monitor the collection canister and stop suction before the liquid reaches the maximum fill line.
9. If suction is weak, the gauge is abnormal, fluid enters the pump, or an alarm occurs, stop use and follow the troubleshooting procedure.
10. After use, switch off the unit and disconnect power before dismantling components for cleaning.
11. Dispose of collected material and single-use accessories according to infection-control and biomedical-waste procedures.
12. Clean and disinfect reusable components as instructed, replace required filters, and document use or faults.

⚠ Safety

- Patient aspiration procedures must be performed only by trained and authorised healthcare personnel under the applicable clinical protocol.
- Never exceed the device's specified vacuum range or an authorised clinical suction setting.
- Do not use damaged tubing, an overfilled canister, or a missing or incorrectly fitted filter.
- Prevent collected fluid from entering the pump; stop immediately if overflow or contamination is suspected.
- Use appropriate PPE and standard precautions when handling secretions or contaminated components.
- Disconnect power before cleaning or servicing and never immerse the electrical pump unit in liquid.

Care & storage

- Empty and clean the collection canister according to the manufacturer's instructions and local infection-control policy.
- Inspect tubing, seals, connectors, filters, and the overflow protection regularly.
- Replace single-use accessories and filters at the specified intervals or when contaminated.
- Check vacuum performance, alarms, and battery condition where applicable.
- Arrange qualified servicing if suction is unreliable, the gauge fails, or fluid enters the pump.

Common mistakes to avoid

- Using the pump without checking the canister, filter, or tubing connections.
- Selecting a suction pressure without following the relevant clinical protocol.
- Allowing the collection canister to overfill.
- Reusing single-use accessories or cleaning the electrical unit by immersing it in liquid.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.